

Homework #1

CS 4342, Fall 2026

100 points total [13.3% of your final grade]

Due: Sept 2, 2026 by 11:59pm

Delivery: Submit via Canvas

For this assignment, you will:

- (70 pts) Implement linear regression with gradient descent
- (30 pts) Make predictions by using your implementation

Part 1: Implement linear regression with gradient descent

In this problem, you will implement the linear regression algorithm in python3. We provide the following files:

- a) `linear_regression.py` - You will implement several functions. As we discussed in class, implement the functions by using **vectorization**.
Do not change the input and the output of the functions.
- b) `test.py` - This file includes unit tests. Run this file by typing `'pytest -v test.py'` in the terminal as you did in homework 1 in order to check whether all of the functions are properly implemented. No modification is required.

Part 2: Make predictions by using your implementation

Given training and test sets, you will make predictions of test examples by using your linear regression implementation (`linear_regression.py`). We provide the following file:

- a) `application.py` – write your code in this file. Do not change `X` and `y`.
- (1) Please play with the hyperparameters `alpha` and number of epochs to make sure your testing loss is smaller than $1e-2$ (i.e., 0.01), and report your hyperparameters, training loss and testing loss.
 - (2) In addition, based on your observations, report a relationship between `alpha` and number of epochs by choosing various `alpha` and number of epochs (i.e., at least four pairs: low and low, low and high, high and low, and high and high `alpha` and number of epochs). Note that a single epoch means the single time you see all examples in the training set.
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What to turn in:

- Submit to Canvas your `linear_regression.py`, `application.py` and a pdf document for part 2.
- This is an individual assignment, but you may discuss general strategies and approaches with other members of the class (refer to the syllabus for details of the homework collaboration policy).